

5(a)(i)	region (of space)	B1
	where a particle experiences a force	B1
5(a)(ii)	similarity – any one point from: • both have an inverse square variation • both decrease with distance • both are radial	B1
	difference – any one point from: • gravitational field always towards (the mass) • electric field can be towards or away from (the charge)	B1
5(b)(i)	$E = Q / 4\pi\epsilon_0 x^2$	C1
	$Q = 4\pi \times 8.85 \times 10^{-12} \times 84 \times 0.15^2$ $= 2.1 \times 10^{-10} \text{ C}$	A1
5(b)(ii)	$E = 84 \times (0.15 / 0.45)^2$ or $E = (2.1 \times 10^{-10}) / (4\pi \times 8.85 \times 10^{-12} \times 0.45^2)$	C1
	$E = 9.3 \text{ V m}^{-1}$	A1
5(c)	line at $E = 0$ from $x = 0$ to $x = 0.15 \text{ m}$	B1
	smooth curve with decreasing negative gradient throughout, from $x = 0.15 \text{ m}$ to $x = 0.45 \text{ m}$, passing through (0.15, 84)	B1
	line passing through (0.45, 9.3)	B1